Schema Renaming

Schema renaming lets you create a version of an existing schema with some component names changed. The notation S[new/old] is per Z RM §3.11.

$$\begin{array}{|l}
\textit{Counter} \\
\hline
\textit{count} : \mathbb{N} \\
\hline
\textit{count} \geq 0 \\
\hline
\end{array}$$

Single rule: produce a Counter with component "count" renamed to "n".

$$\textit{CounterN} \mathrel{\widehat{=}} \textit{Counter}[n/\textit{count}]$$

Multiple renames in one step:

$$\begin{array}{|l}
\textit{Point} \\
\hline
x : \mathbb{N} \\
y : \mathbb{N} \\
\hline
\end{array}$$

Rename both components at once.

$$\textit{SwappedPoint} \mathrel{\widehat{=}} \textit{Point}[x/y, y/x]$$

Renaming Primed Schemas

The decoration is tightest-binding per Z RM §3.11: S'[a/b] renames the primed schema S', not the result of renaming then priming.

$$\textit{Op}2 \mathrel{\widehat{=}} \textit{Counter}[\textit{count}'/\textit{count}]$$

In an operation schema context, this captures the after-state renaming: the existing component count is renamed to the primed component count' in the view.

Decorated Component Names

The rename pairs themselves may carry decoration.

$$\textit{CounterOp} \mathrel{\widehat{=}} \textit{Counter}[\textit{count}'/\textit{count}]$$